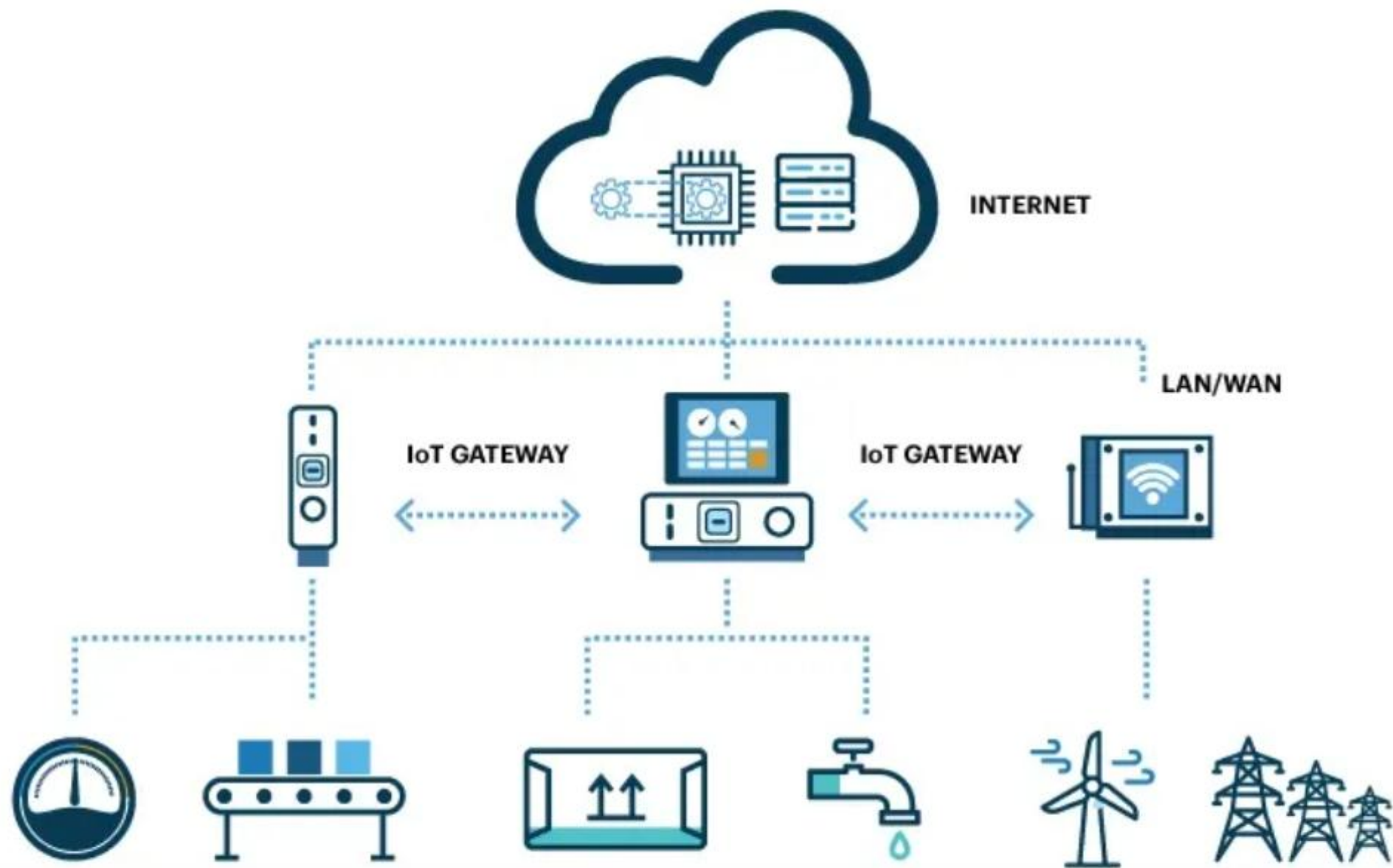


IoT & Edge Computing

Dr. Shubhangi Deshmukh



IoT Architectures

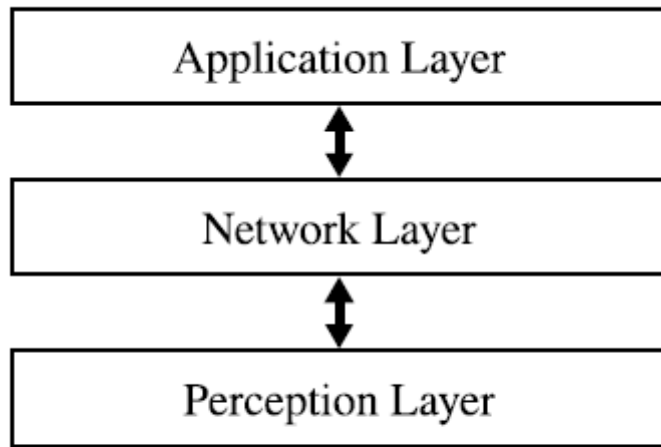
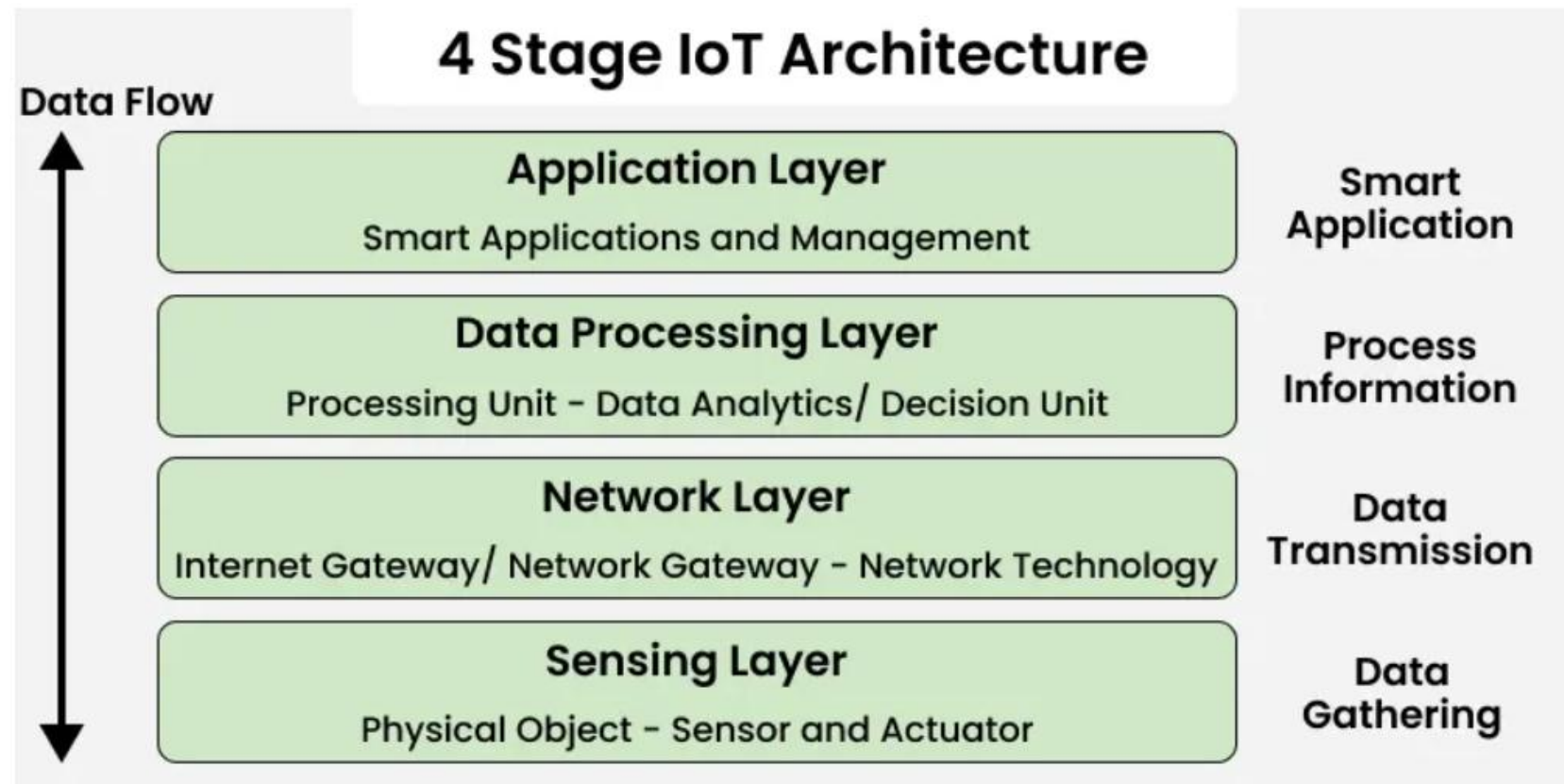


Fig. 1. IoT three-layer architecture.



Architecture of IoT

Service Oriented Architecture (SOA based IoT architecture)

Model	Best for	Pros	Cons
3-layer	Small IoT pilots	Simplest design; fast to deploy	Limited separation of concerns; scaling constraints
5-layer	Growing deployments	Balanced modularity; clearer data flow	Moderate complexity
7-layer	Enterprise/IIoT scale	Strong isolation; governance and security clarity	Highest complexity; requires mature ops

IoT Reference Architecture

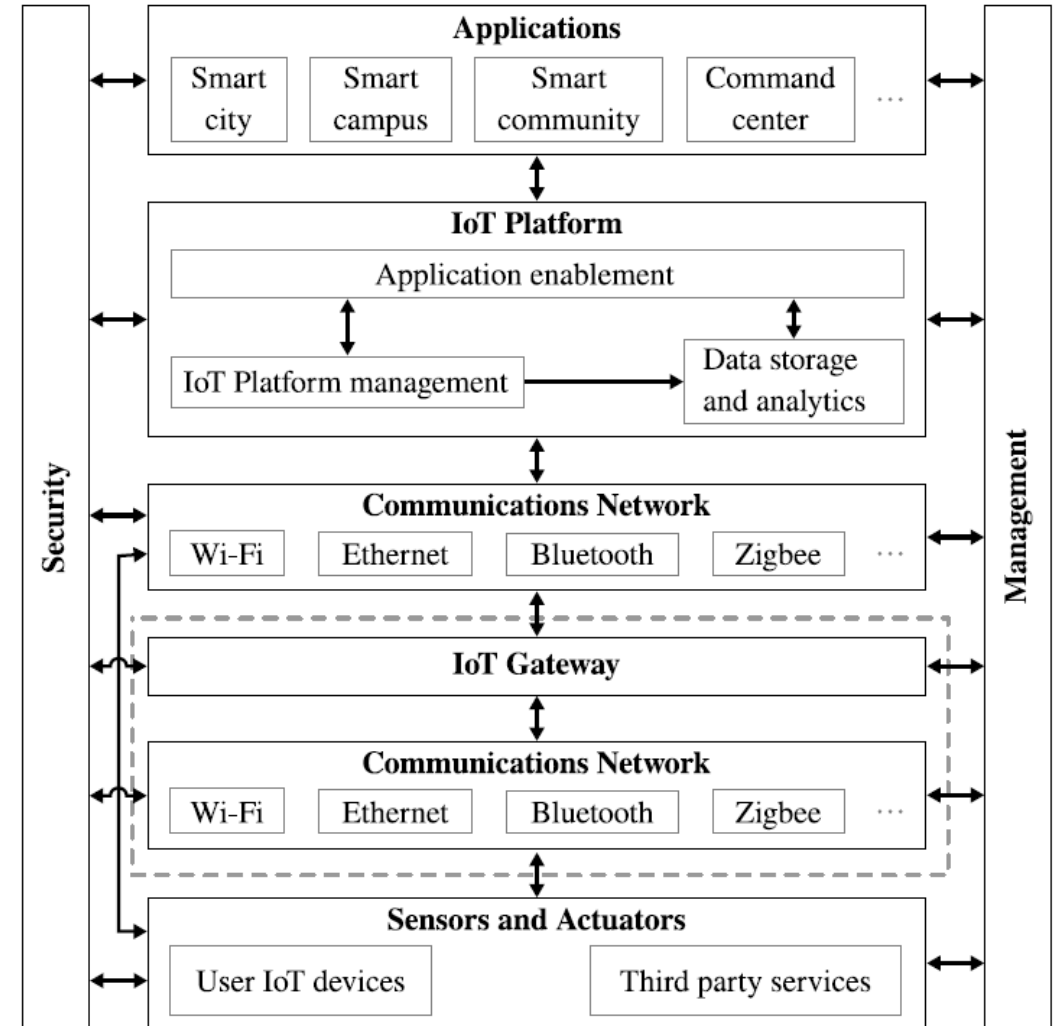
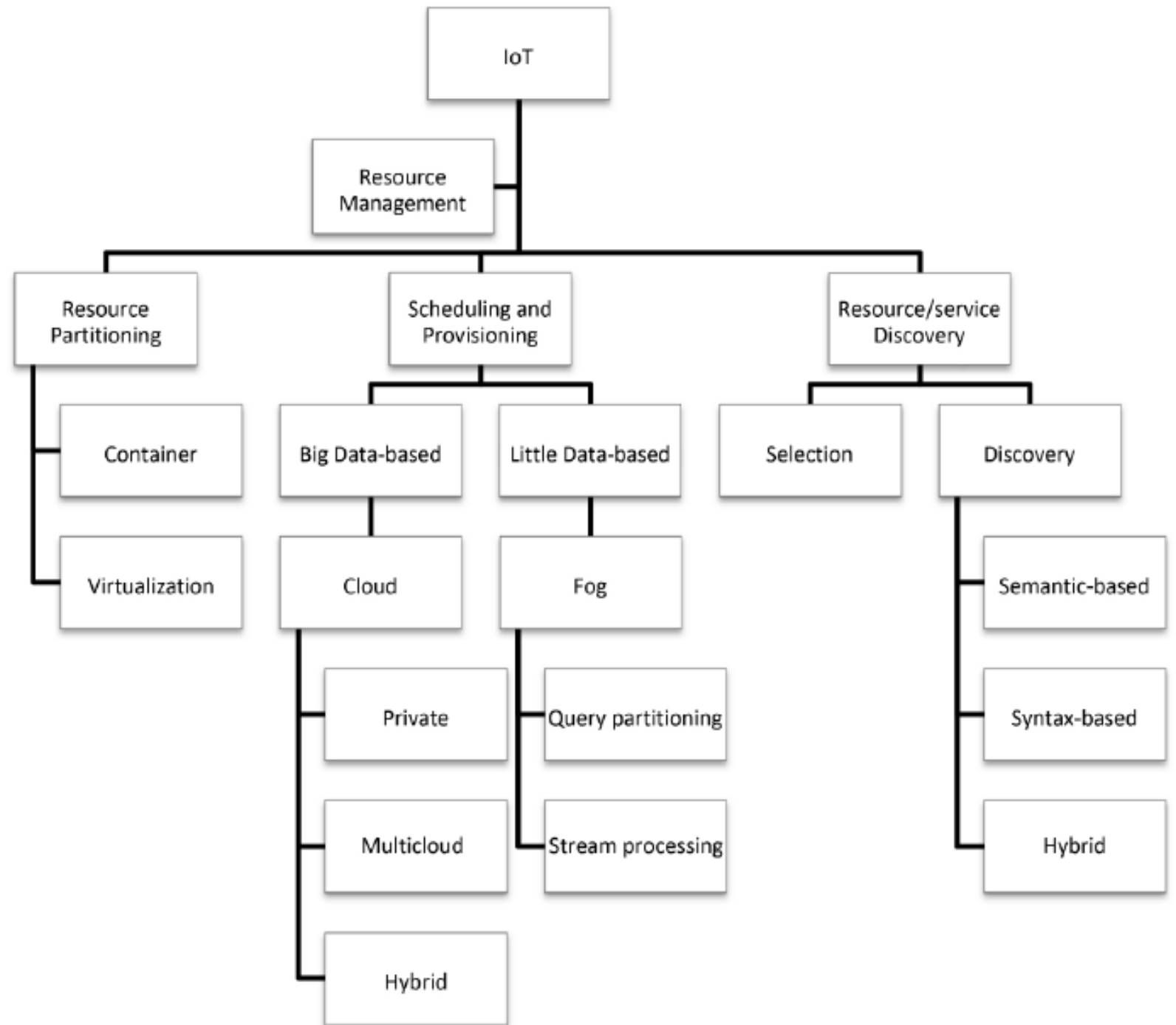


Fig. 3. IoT architecture.

Resource management in IoT



Cloud Service Models for IoT

1

Infrastructure as a Service:

Full Control for Advanced Customization

2

Platform as a Service:

Flexibility for Developers

3

Software as a Service:

Simplified and Ready-Made Solutions

	IaaS	PaaS	SaaS
Use cases	Virtual machines, custom gateways, storage solutions like AWS and GCP	Real-time data processing platforms, development and deployment environments, automation solutions like 2Smart Standalone	Data analytics, device management platforms like 2Smart Cloud and Propuskator
Control	Full control over hardware, OS, and configurations	Control over apps and data, limited control over runtime and OS	No control over infrastructure or backend processes
Scalability	Highly scalable based on user configurations	Highly scalable based on project requirements	Scales automatically with minimal user input
Maintenance	Requires dedicated maintenance	Maintenance on the provider's side	Maintenance on the provider's side
Cost-effectiveness	Potentially high costs depending on usage and scaling	Typically more cost-efficient than IaaS	Cost-effective, subscription-based pricing
Flexibility	Maximum flexibility for custom solutions	Flexible for app development yet limited by platform constraints	Limited flexibility, best for ready-to-use solutions

Criteria	SaaS (Microsoft Azure IoT Central)	PaaS (AWS IoT Core)	IaaS (Google Cloud Compute Engine, BigQuery)
Cost	Low — fixed subscription or pay-per-use	Medium — pay-per-use of resources	High — pay-per-use of resources and their usage
Control	Minimal — users only manage settings and analytics	Medium — control over app development, not infrastructure	Maximum — full control over infrastructure and configuration
Scalability	Limited — depends on platform capabilities	High — ability to integrate with other services	Highest — full customizability and resource selection
Deployment speed	High — quick setup without the need for development	Medium — requires development, but ready-made services speed up the process	Low — requires server setup, security, and service deployment

IaaS Solutions

Virtual machines for IoT data processing.

You can configure virtual machines to collect, process, and store large amounts of data.

Custom IoT gateways.

You may require more specific connectivity options than Bluetooth or WiFi in IoT projects for gateways that support IaaS.

Storage solutions.

If your project generates huge amounts of data, IaaS lets you implement your own storage

PaaS Solutions

Data processing platforms.

You can leverage real-time data processing services to analyse large amounts of IoT data.

Development and deployment environments.

Platforms that support development and rapid deployment can seep up your IoT project by providing the necessary libraries, SDKs, and APIs.

SaaS Solutions

IoT data analytics.

Many companies offer an IoT SaaS solution for collecting, visualizing, and analysing data.

Device management platforms.

You can use SaaS solutions for device monitoring, diagnostics, and updates to manage devices at scale without developing these capabilities from scratch.

Platform as a Service



PaaS is best for companies looking to custom build a solution.

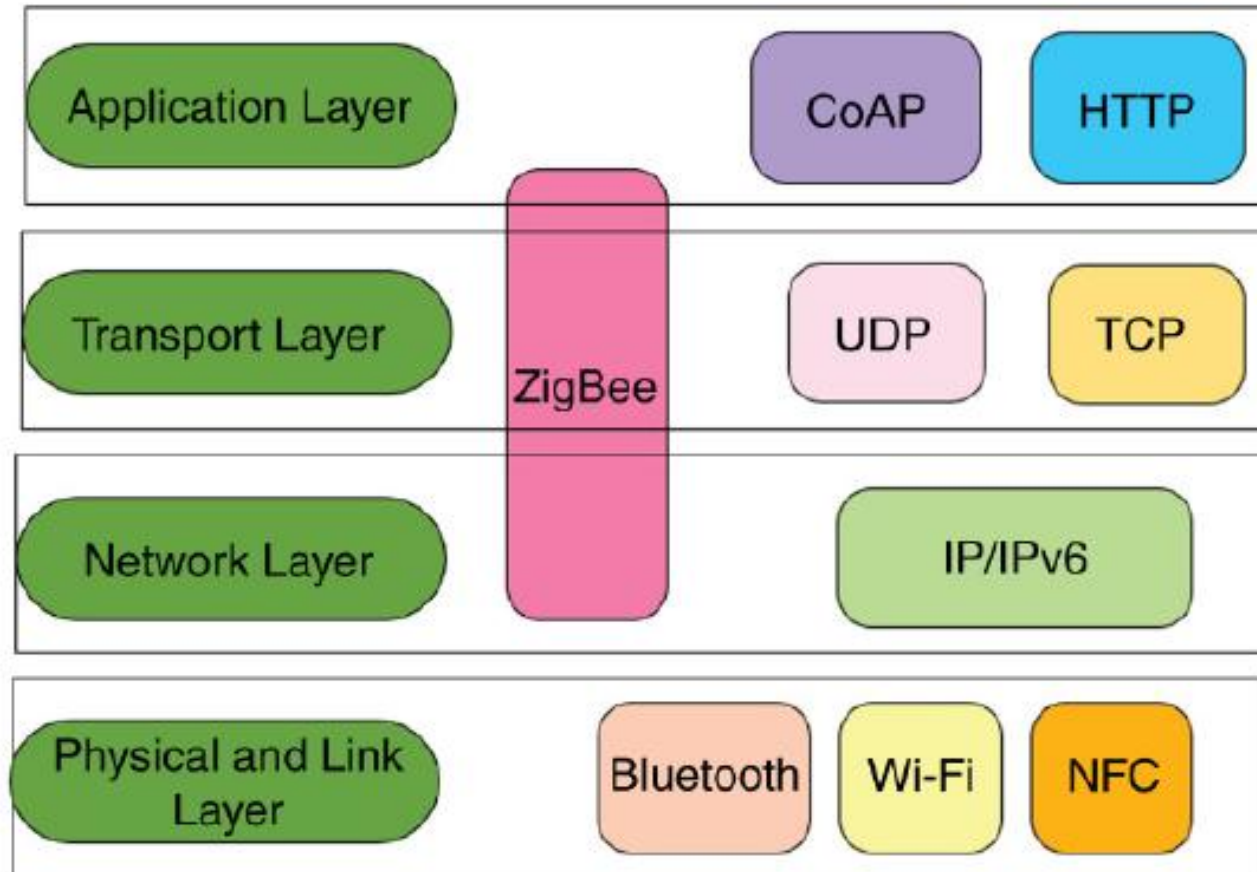
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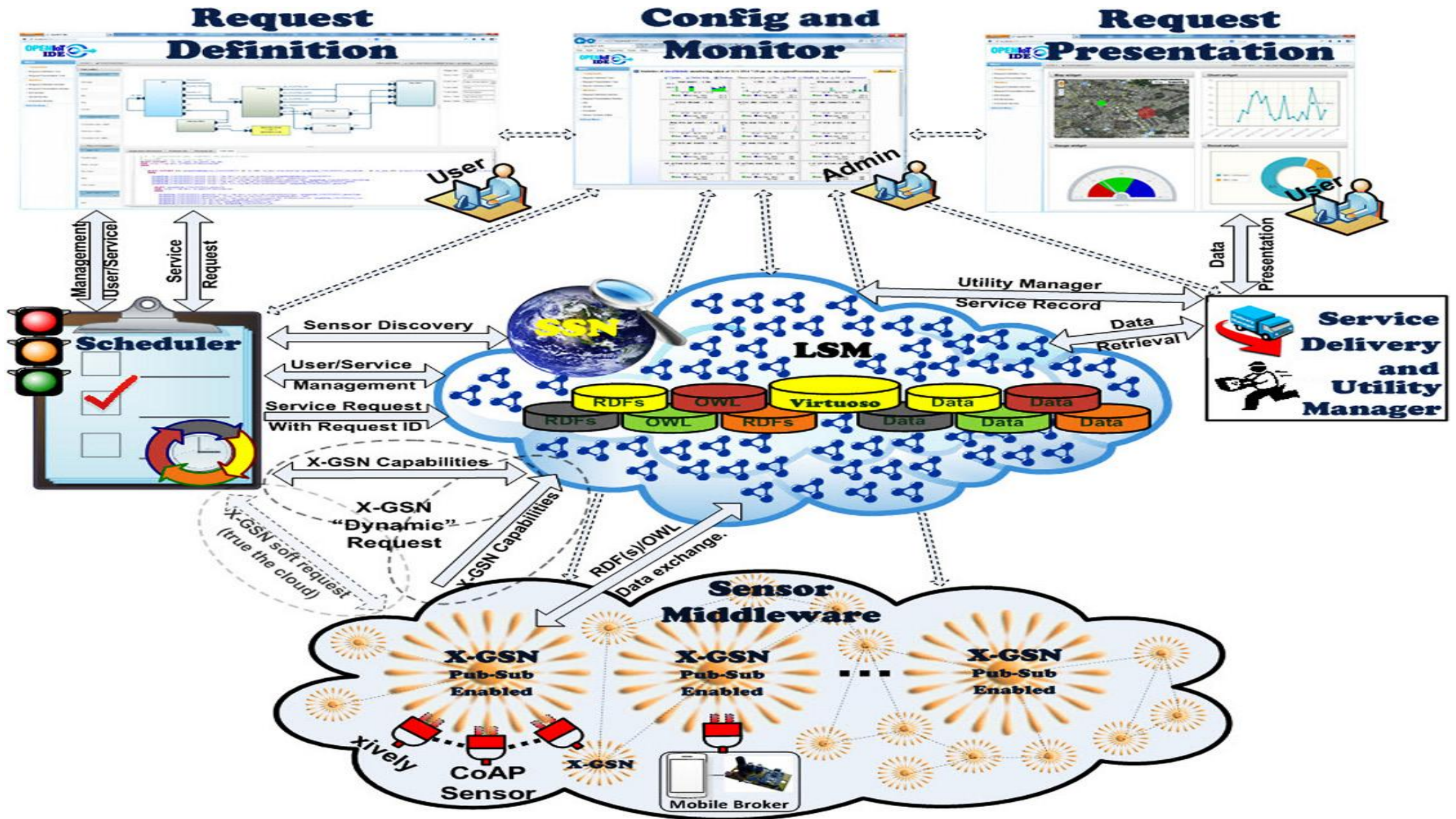
Software as a Service



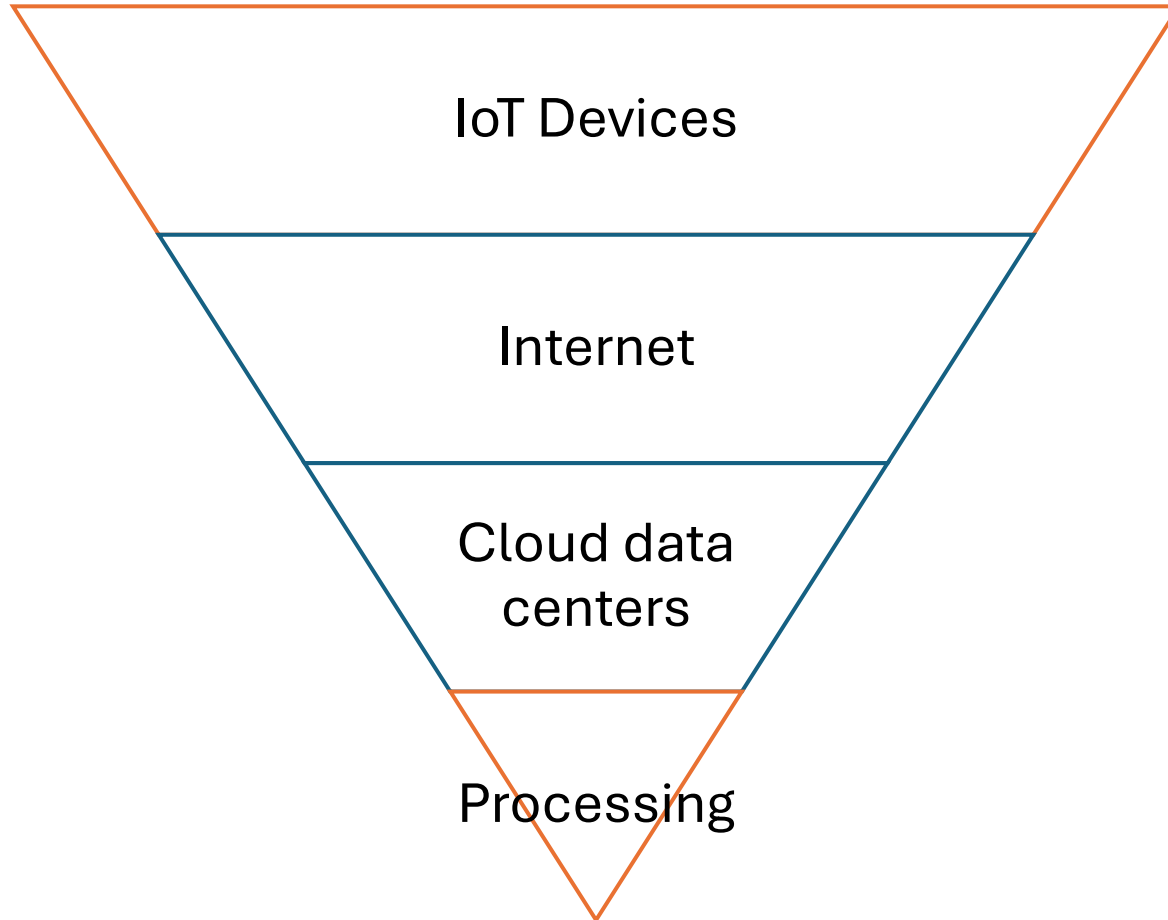
SaaS is best for companies looking for an out-of-the-box ease of use.

Various Protocols in IoT Applications





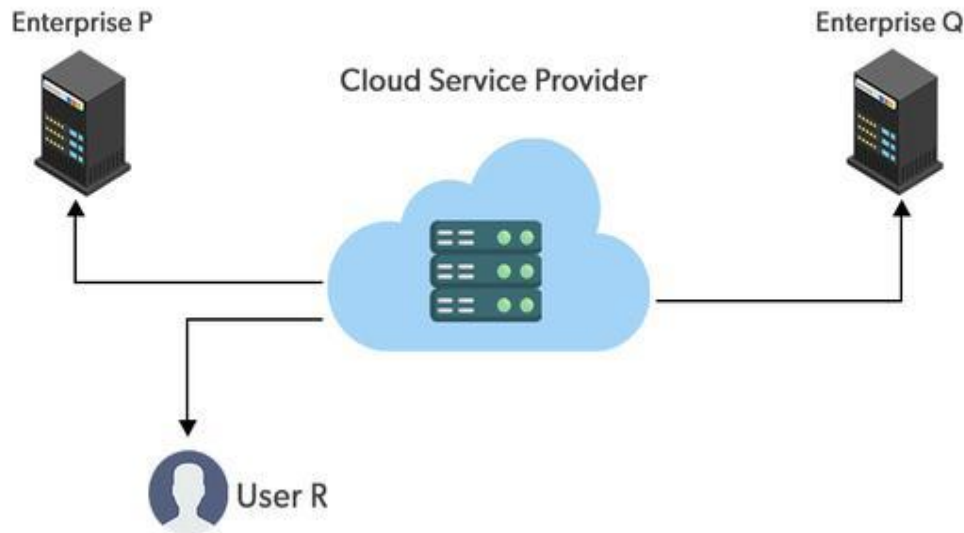
Traditional cloud model



Disadvantages

- High latency
- Network congestion
- Large bandwidth consumption
- Privacy concerns
- Slow decision making

Public cloud

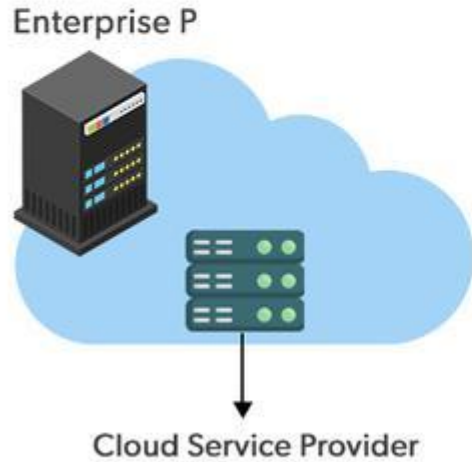


- Public cloud is easily able to scale up and down resources as per the demand of traffic and workload. It facilitates with performance optimization and cost efficiency.
- It works on pay-as-you-go cloud model and helps in resolving the investments needs in hardware and infrastructure reducing overall costs.
- It is difficult to trust and maintain data to a third-party provider may raise concerns about control and ownership
- The shared infrastructure of public cloud resources increases the risk of data breaches and unauthorized access. It raises security and privacy concerns.
- Public cloud comes with limited transparency about the underlying infrastructure which may make it challenging to monitor and manage performance effectively.

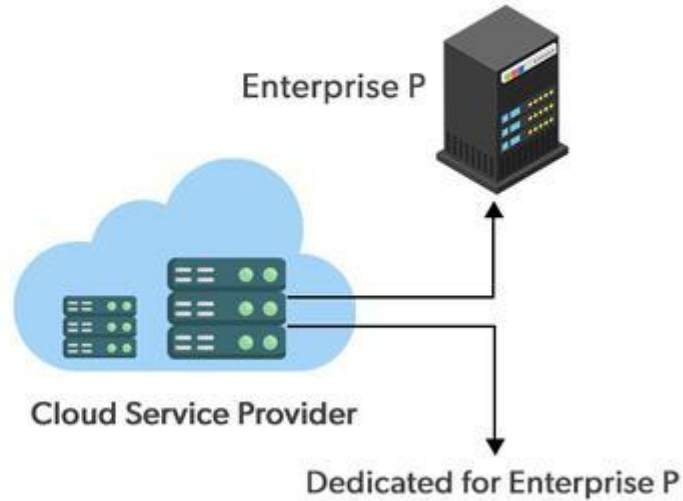
Examples: Amazon_EC2, IBM, Azure, GCP(Google Cloud Platform)

Private Cloud

On premise Private cloud



Externally hosted Private cloud



Examples:

VMware vCloud Suite, OpenStack, Cisco Secure Cloud, Dell Cloud Solutions, HP Helion Eucalyptus

Private cloud providers are HP Data Centers, Ubuntu, Elastic-Private cloud, Microsoft, etc.

Customer information protection: In the private cloud security concerns are less since customer data and other sensitive information do not flow out of private infrastructure.

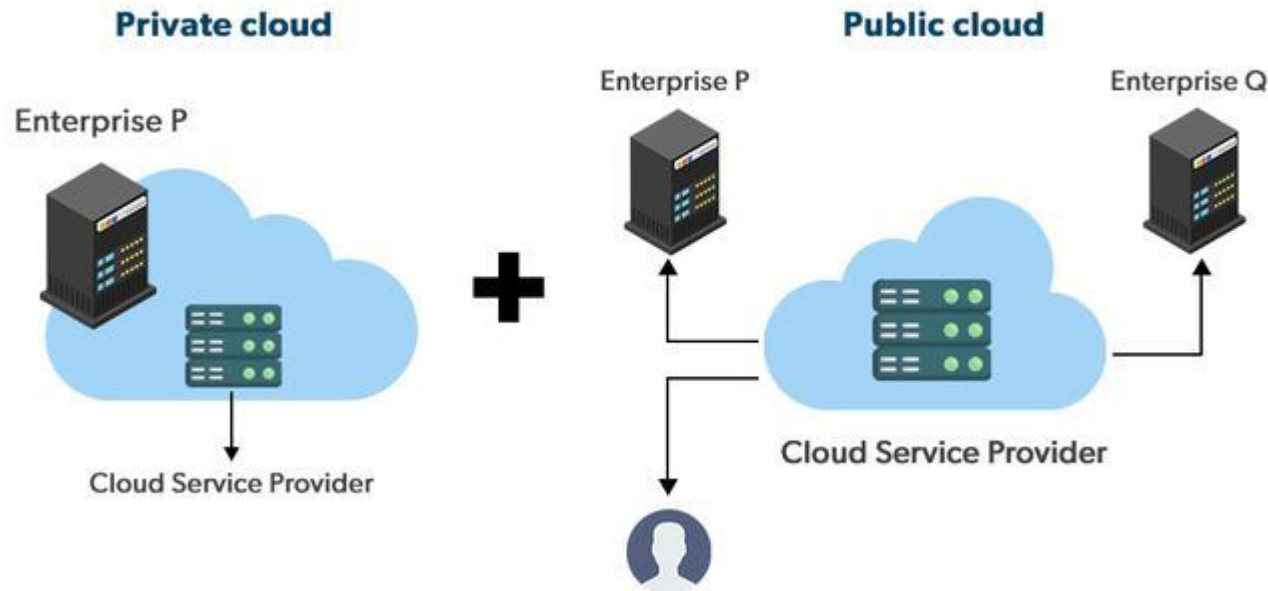
Infrastructure ensuring SLAs: Private cloud provides specific operations such as appropriate clustering, data replication, system monitoring, and maintenance, disaster recovery, and other uptime services.

Compliance with standard procedures and operations: Specific procedures have to be put in place when deploying and executing applications according to third-party compliance standards. This is not possible in the case of the public cloud.

The restricted area of operations: Private cloud is accessible within a particular area. So the area of accessibility is restricted.

Expertise requires: In the private cloud security concerns are less since customer data and other sensitive information do not flow out of private infrastructure. Hence skilled people are required to manage & operate cloud services.

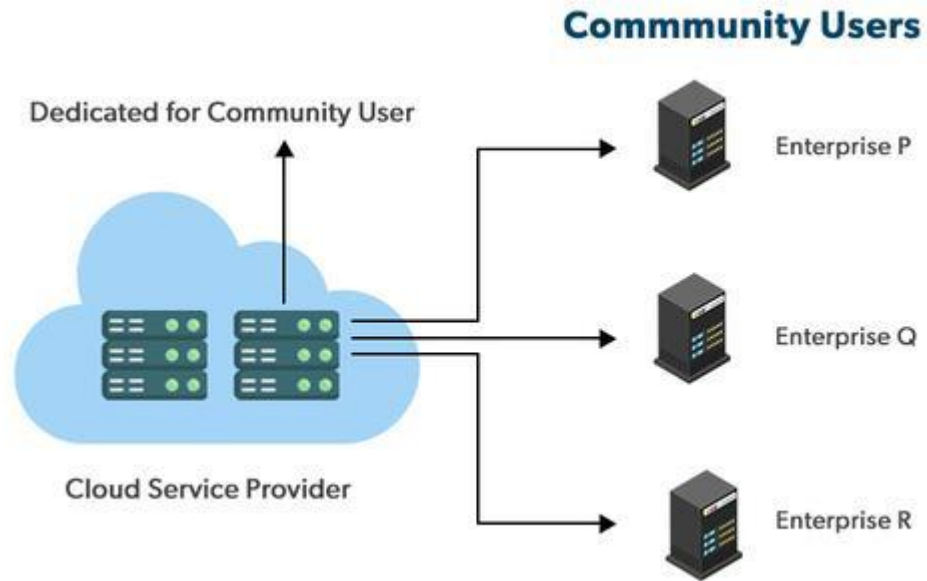
Hybrid Cloud



Examples: AWS Outposts, Azure Stack, Google Anthos, IBM Cloud Satellite, Oracle Cloud at Customer

- Hybrid cloud is available at a cheap cost than other clouds because it is formed by a distributed system.
- It works comes up with working fast with lower cost and facilitates in reducing the latency of the data transfer process.
- Most important thing is security. A hybrid cloud is totally safe and secure because it works on the distributed system network.
- It's possible that businesses lack the internal knowledge necessary to create such a hybrid environment. Managing security may also be more challenging. Different access levels and security considerations may apply in each environment.
- Managing a hybrid cloud may be more difficult. With all of the alternatives and choices available today, not to mention the new PaaS components and technologies that will be released every day going forward, public cloud and migration to public cloud are already complicated enough. It could just feel like a step too far to include hybrid.

Community cloud

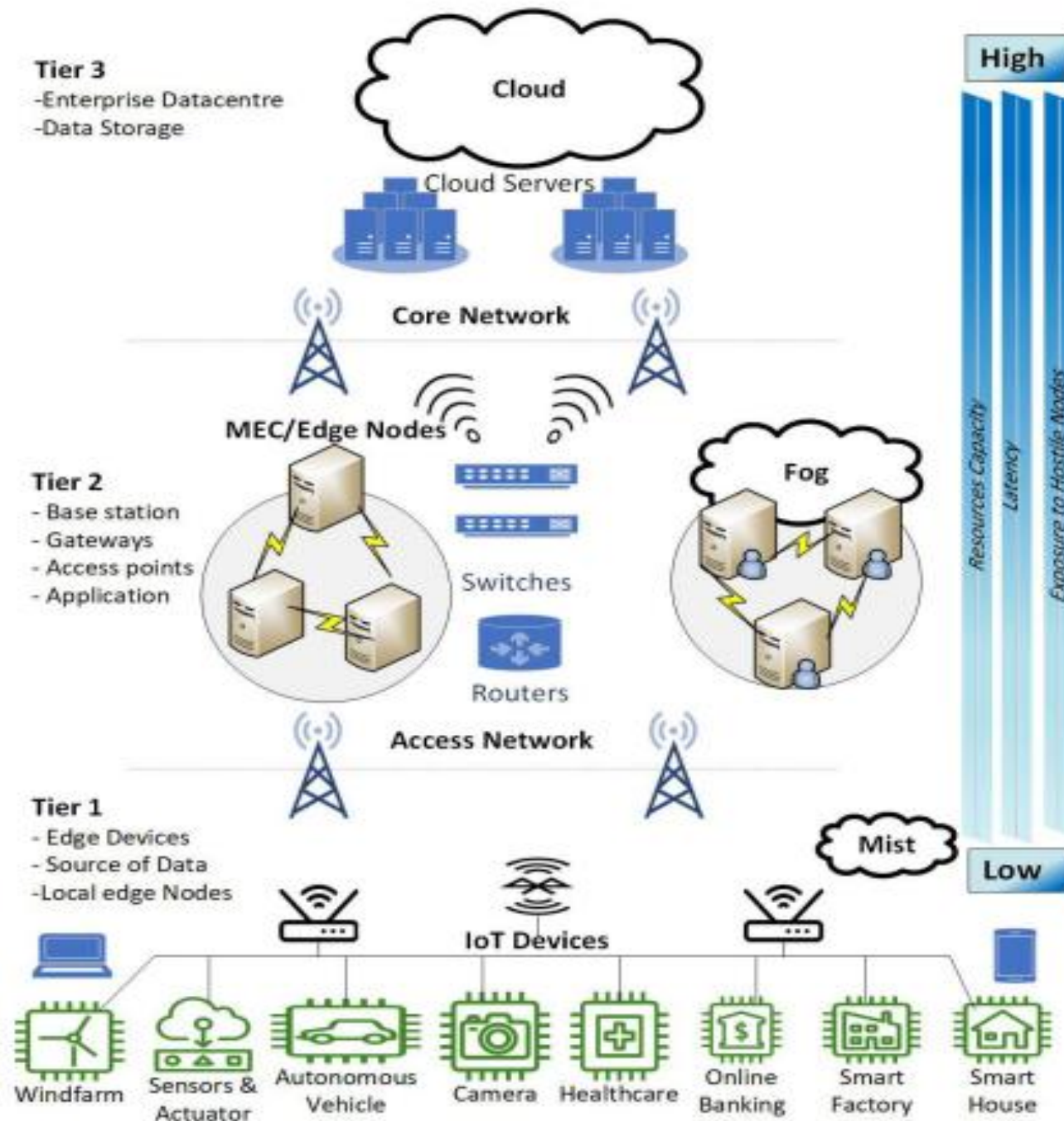


Examples: CloudSigma, Nextcloud, Synology C2, OwnCloud, Stratoscale

Multicloud

Examples: Cloud Foundry, Kubernetes, Apache Mesos, Red Hat OpenShift, Docker Swarm

Aspect	Public Cloud	Private Cloud	Hybrid Cloud
Infrastructure	Shared among multiple distinct organizations.	Dedicated exclusively to a single organization.	A customized mix of shared and dedicated resources.
Cost Model	Pay-as-you-go; minimal upfront costs.	High initial investment and ongoing maintenance.	Variable; optimizes costs by routing workloads efficiently.
Control & Security	Managed by the provider; lower control for the user.	Full internal control; highest level of security.	Shared responsibility; requires complex security routing.
Flexibility	High resource flexibility, low configuration flexibility.	High configuration flexibility, restricted scaling capability.	Ultimate flexibility leveraging the best of both worlds.

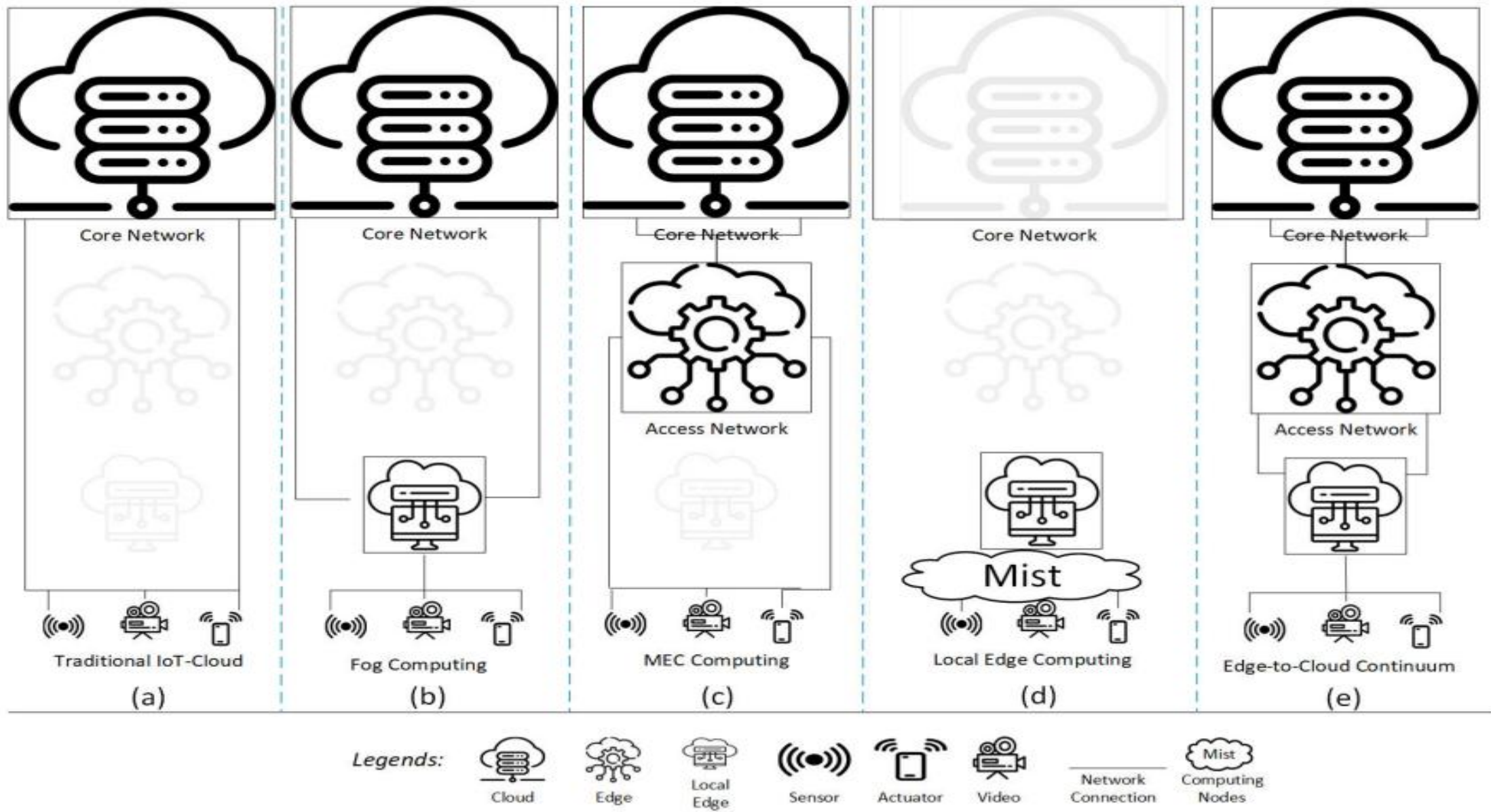


Three-tier edge–cloud continuum:

Tier 1—IoT devices and local Edge Computing;

Tier 2—Edge Computing and Fog Computing nodes;

Tier 3—Cloud data centres.



IoT service orchestration to computing nodes: (a) centralized IoT-CC; (b) FC; (c) multiaccess EC; (d) local EC; and (e) three-tier edge-cloud continuum.

Edge computing Architectures and Technology

- **Cloudlet Computing.** This refers to computing resources (small cluster) connected via WLAN to the end-users. In general, it can be considered as a "data centre in a box" which provides support (computing and storage) to the end-users over the WLAN network. Cloudlet Computing is based on three layers: the component layer, the node layer, and the cloudlet layer. This is designed to have higher bandwidth, thus lowering the latency for the applications.
- **Fog Computing-** a decentralized computing resource that can be placed anywhere between the cloud and the end-users. It is based on the so-called Fog Computing Nodes (FCNs) [14]. All of these FCNs are heterogeneous, including switches, routers, and access points. FCNs heterogeneous environment facilitates the devices at different protocol layers and non-IP based technologies to communicate between the FCNs and the end device. These FCNs are hidden for the end-users, thus ensuring security.
- **Multi-access Edge Computing (MEC),** which refers to implementing Edge Computing within the Radio Access Network to reduce the latency. Formally known as Mobile Edge Computing, it is an ETSI-defined network architecture located closer to the Radio Network Controller or macro base station. The edge orchestrator organises the MEC, provides network information about load and capacity, and owners information to the end-users about their location and network information.
- **IoT (Internet of Things)** contains a large set of devices and sensors that produce a huge volume of data. These also exchange the data through a modern communication network and monitor and control the infrastructure. Typically, end-users at the Edge use the IoT devices and sensors.

Feature/Aspect	Traditional IoT-cloud	Fog computing	MEC	Local edge computing	Three-tier computing continuum
Key features	Ensure effective, scalable resource availability for extensive IoT data.	Data processing proximity to user, with low latency communications.	Processing, storage and data computation to the edge of the network.	Enhance data privacy and security. Decision making and processing data locally.	Integrated computing continuum, and enhance QoS performance.
Strong aspects	Well-recognized tools and frameworks for QoS requirements of IoT.	Instant response to time-critical applications, with scalable resources.	Support high bandwidth and ultra-low latency for IoT applications.	Instant data processing for real-time applications, control over data.	Distribute workload in three-tier, ensuring continued application performance.
Weak aspects	Challenges of high latency and connectivity for time-critical applications.	Resource management and security challenges of fog nodes.	Limited resources, challenges of defining orchestration protocols.	Challenges of limited computing resources for massive IoT.	Increased complexity, challenges in adaptive offloading.

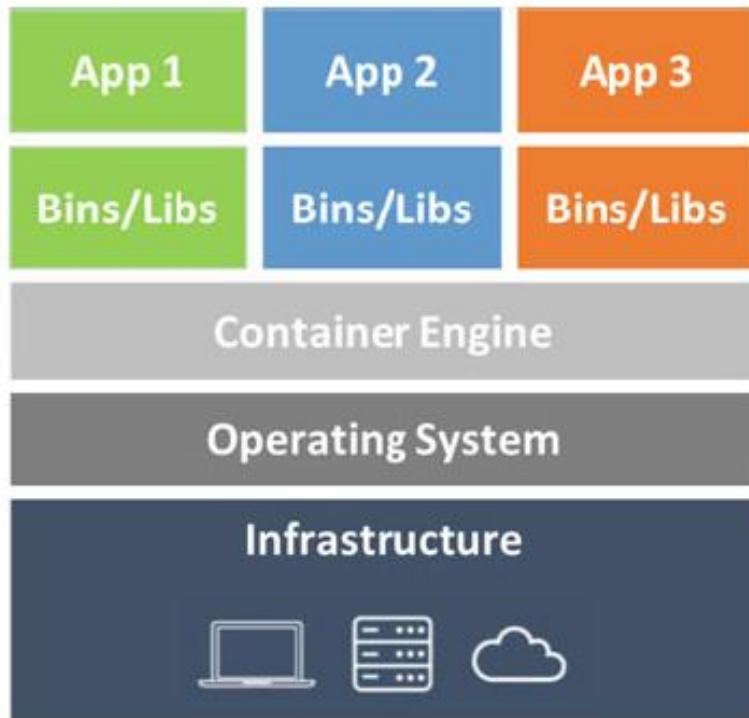
Virtualisation

- In Computing - virtualisation often implies the reference to the creation of a Virtual Machine (VM) managed by a hypervisor, a middleware responsible for providing an abstraction or emulation layer from the hardware.
- The hardware running the hypervisor is called the host whereas all emulated VMs running inside them are referred to as guests.
- Type-1 (Native) hypervisors are running directly on hardware (hence often referred to as bare metal hypervisors).

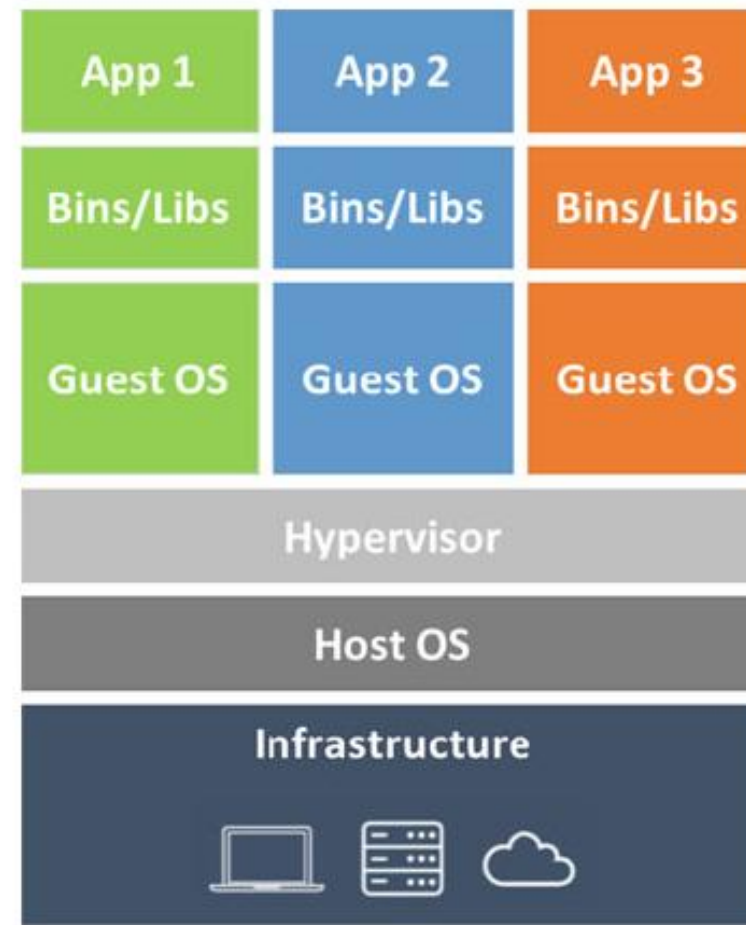
Examples- Xen , VMware ESXI or Hyper-V

- Type-2 (Hosted) hypervisors require a host operating system whose capabilities are used in order to perform virtualization operations and emulations.

Examples- KVM or VirtualBox



(a) Container Environments [7].



(b) Type-1 (Bare-Metal) or Type-2 (Hosted) Hypervisors [7].

Kubernetes and provides fundamental infrastructure support for network, application deployment and metadata synchronization between Cloud and Edge architectures.

Requirements of Edge device management & orchestration

Requirement	Description
Scalability	Ability to address a large number of Edge devices of different type and capabilities with appropriate deployment and communicating protocol.
Security	Privacy preserving for security tokens and support for integrity checks within the infrastructure
Heterogeneity	Support for a high degree of heterogeneity within hardware/software
Volatility	Support for volatile availability and mobile hardware/software components
Data Protection	GDPR Compliance, ensure all data is kept locally and on-the-fly encrypted
Infrastructure Performance	Very low latency, lightweight publish-subscribe network protocol as MQTT [27]. High performance containerised resources with fast (Zero-touch) provisioning allowing easy system upgrades
Application Portability	Unified architecture view via MEC compliance enabling Function as a Service (FaaS) capabilities
Data Analytics	Supports for Data Management and Data Analytics Pipeline Engine.

